

Information Field Theory

And Climatology / Geophysics

IFT-Climate v1.0

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The application of Information Field Theory (IFT) to climatology and geophysics is presented. The Earth's climate system is modeled as an informational field $\rho(x, t)$ where x represents geographic coordinates and thermodynamic variables. The global temperature emerges as the informational gradient $T \sim \Xi = |\nabla \ln \rho|$. Climatic tipping points (collapse of the AMOC, permafrost thaw, Amazon dieback) are interpreted as informational phase transitions: when the product $\tau\Xi$ falls below a critical threshold K_{climate} , the subsystem loses coherence and transitions to a new state. Oceanic circulation (AMOC, Gulf Stream) is described as an informational flow governed by $\partial_t \rho = \nabla \cdot (D\rho \nabla \delta \mathcal{F} / \delta \rho)$. Recent observational data (2020–2025) are analyzed under this framework, and early-warning indicators based on the temporal evolution of Ξ and τ are proposed.

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Chapter 1

Introduction: The Climate as an Informational System

1.1 Motivation

The Earth's climate system is a complex, nonlinear system with multiple coupled components (atmosphere, oceans, cryosphere, biosphere). Traditional climate models (GCMs) are computationally expensive and have difficulty predicting tipping points. IFT offers a complementary framework based on information: modeling the climate as an informational field ρ where critical transitions emerge from the geometry of the field.

1.2 Central Thesis

Theorem 1.2.1 (IFT-Climate Thesis). *The Earth's climate is a coupled informational system. Climatic variables (temperature, pressure, salinity) are manifestations of the field ρ , and abrupt climatic changes (tipping points) are informational phase transitions characterized by $\tau\Xi < K_{\text{climate}}$.*

Chapter 2

The Climate System as an Informational Field

2.1 Definition of ρ for the Climate

$$\rho(\mathbf{x}, t) = \frac{E(\mathbf{x}, t)}{\int E(\mathbf{x}', t) d\mathbf{x}'}, \quad (2.1)$$

where $E(\mathbf{x}, t)$ is the energy density at the geographic coordinate $\mathbf{x}$ (including temperature, humidity, wind kinetic energy). ρ represents the spatial distribution of climatic energy.

2.2 Climatic Informational Gradient

$$\Xi(\mathbf{x}, t) = |\nabla \ln \rho(\mathbf{x}, t)|. \quad (2.2)$$

Ξ measures the intensity of climatic gradients: fronts, ocean currents, convergence zones. Where Ξ is high, there are strong thermal contrasts and intense dynamics.

2.3 Global Temperature as Average Gradient

IFT predicts that the global mean temperature is related to the average informational gradient:

$$\bar{T}(t) \propto \langle \Xi \rangle(t) = \int \rho(\mathbf{x}, t) \Xi(\mathbf{x}, t) d\mathbf{x}. \quad (2.3)$$

An increase in $\langle \Xi \rangle$ corresponds to an increase in global temperature (greater energy contrast = higher mean temperature).

Chapter 3

Tipping Points as Informational Phase Transitions

3.1 Climatic Stability Criterion

A climatic subsystem (e.g., AMOC, Amazon, permafrost) is stable if:

$$\tau_{\text{sub}} \cdot \Xi_{\text{sub}} > K_{\text{climate}}, \quad (3.1)$$

where τ_{sub} is the characteristic persistence time of the subsystem and Ξ_{sub} is its internal informational gradient.

3.2 Tipping Point Mechanism

When $\tau\Xi$ approaches K_{climate} from above:

1. Fluctuations are amplified (critical slowing down).
2. The variance of Ξ increases (loss of stability).
3. Upon crossing the threshold, the system transitions to a new state with a different ρ .
4. The transition is irreversible if the new state is separated by a free-energy barrier.

3.3 Examples of Tipping Points

Subsystem	Ξ_{sub}	τ_{sub}	Current Status
AMOC (Atlantic circulation)	High (strong T/S gradient)	~ 100 years	Approaching threshold
Amazon (biome)	Medium (moisture gradient)	~ 50 years	At risk
Permafrost (frozen soil)	Low (thermal gradient)	~ 1000 years	Thawing
Arctic sea ice	High (albedo)	~ 10 years	Collapsing
Greenland (ice sheet)	Medium	~ 1000 years	Retreating

Chapter 4

Oceanic Circulation as Informational Flow

4.1 AMOC as a Flow of ρ

The Atlantic Meridional Overturning Circulation (AMOC) is described as an informational flow:

$$\partial_t \rho = \nabla \cdot \left(D(\rho) \rho \nabla \frac{\delta \mathcal{F}}{\delta \rho} \right), \quad (4.1)$$

where $\mathcal{F}[\rho] = E[\rho] - T_{\text{ocean}} S[\rho]$ is the free energy of the ocean-atmosphere system.

4.2 Collapse of the AMOC

The collapse of the AMOC occurs when the informational flow is interrupted ($D(\rho) \rightarrow 0$ in certain regions), creating a barrier that prevents northward heat transfer. In IFT terms:

$$\tau_{\text{AMOC}} \cdot \Xi_{\text{AMOC}} \rightarrow K_{\text{climate}}^-. \quad (4.2)$$

4.3 Early-Warning Indicator

The product $\tau(t) \cdot \Xi(t)$ measured in the North Atlantic can serve as an early-warning indicator. Recent observational data (2020–2025) suggest that this product is decreasing, indicating approach to a tipping point.

Chapter 5

Analysis of Recent Observational Data (2020–2025)

5.1 Global Temperature

The 2023–2024 period recorded record global temperatures ($\sim 1.5^\circ\text{C}$ above preindustrial levels). In IFT terms, this corresponds to an increase in the global $\langle \Xi \rangle$.

5.2 Extreme Events

The frequency of extreme events (heat waves, floods, hurricanes) has increased. IFT interprets this as an increase in the variance of Ξ : the informational field becomes more “rugged”, with intense localized gradients.

5.3 AMOC Slowdown

Recent studies (2023–2025) indicate that the AMOC is at its weakest point in 1000 years. IFT predicts that $\tau_{\text{AMOC}} \cdot \Xi_{\text{AMOC}}$ is decreasing, approaching the critical threshold.

Chapter 6

Predictions and Recommendations

6.1 IFT Predictions

1. If $\tau_{\text{AMOC}} \cdot \Xi_{\text{AMOC}} < K_{\text{climate}}$ before 2050, the AMOC will collapse with severe climatic consequences (North Atlantic cooling, monsoon displacement).
2. Permafrost thaw will follow a phase-transition dynamics: slow at first, accelerating as the threshold is approached.
3. The Amazon may transition to savanna if $\tau\Xi$ falls below K_{climate} due to deforestation and climate change.

6.2 IFT Indicators for Climate Policy

1. **Climate Stability Index:** $\tau\Xi$ measured for each critical subsystem.
2. **Early Warning:** Continuous monitoring of Ξ and τ from satellite and oceanographic data.
3. **Action Thresholds:** Define K_{climate} for each subsystem and activate policies when $\tau\Xi$ approaches the threshold.

Chapter 7

Master Formula of IFT-Climate

$$\rho(\mathbf{x}, t) = \frac{E(\mathbf{x}, t)}{\int E d\mathbf{x}} \quad (\text{Normalized energy density}), \quad (7.1)$$

$$\Xi(\mathbf{x}, t) = |\nabla \ln \rho| \quad (\text{Climatic gradient}), \quad (7.2)$$

$$\bar{T}(t) \propto \langle \Xi \rangle(t) \quad (\text{Global temperature}), \quad (7.3)$$

$$\partial_t \rho = \nabla \cdot (D \rho \nabla \frac{\delta \mathcal{F}}{\delta \rho}) \quad (\text{Climatic flow}), \quad (7.4)$$

$$\text{Tipping point: } \tau \Xi < K_{\text{climate}} \quad (\text{Coherence collapse}). \quad (7.5)$$

Chapter 8

Final Declaration

IFT-Climate provides a unified framework for understanding the climate system as an informational field. Climatic tipping points emerge naturally as phase transitions when $\tau\Xi < K_{\text{climate}}$. The framework offers quantitative early-warning indicators and can complement traditional climate models.

$$\boxed{\text{Climate} = \text{Energetic informational field } \rho(\mathbf{x}, t)}.$$

$$\boxed{\text{Tipping point} = \text{Phase transition: } \tau\Xi < K_{\text{climate}}}.$$

$$\boxed{\text{Oceanic circulation} = \text{Informational flow}}.$$

8.1 Affirmations

1. The climate is a coupled informational system described by the field $\rho(\mathbf{x}, t)$.
2. Global temperature is proportional to the average informational gradient $\langle \Xi \rangle$.
3. Tipping points are phase transitions when $\tau\Xi < K_{\text{climate}}$.
4. Oceanic circulation is an informational flow governed by the gradient of $\mathcal{F}$.
5. $\tau\Xi$ is an early-warning indicator for climatic collapses.

8.2 Open Problems

1. Calibrate K_{climate} for each subsystem using historical and paleoclimatic data.
2. Estimate τ and Ξ in real time from satellite and oceanographic data.
3. Integrate the IFT framework with GCM models for hybrid predictions.
4. Validate the prediction of increased Ξ variance before tipping points.
5. Extend the analysis to other subsystems (monsoons, ENSO, Antarctic ice).